

ENAUTO Lab Reference - Commands and Why We Use Them

This reference covers every major command used during the initial Raspberry Pi setup and the first Python lesson.

System Updates

Command	Purpose
<code>sudo apt update</code>	Refreshes the package index so the Pi knows about the latest software versions.
<code>sudo apt full-upgrade -y</code>	Installs all available updates, including dependency changes. Keeps the system fully current.
<code>sudo reboot</code>	Restarts the system after major updates.

System Information

Command	Purpose
<code>hostnamectl</code>	Displays hostname, operating system, kernel, and architecture.
<code>free -h</code>	Shows RAM and swap usage in a human-readable format.
<code>df -h</code>	Shows disk usage and free space.
<code>ip addr</code>	Displays network interfaces, IP addresses, and interface status.

SSH and Keys

Command	Purpose
<code>dir ~/.ssh</code> (Windows PowerShell)	Shows existing SSH keys on the Windows PC.
<code>type \$env:USERPROFILE\.ssh\id_ed25519.pub</code>	Displays the public SSH key so it can be copied to a server.
<code>mkdir -p ~/.ssh</code>	Creates the SSH configuration directory if it does not already exist.
<code>nano ~/.ssh/authorized_keys</code>	Stores trusted public keys for passwordless SSH login.
<code>chmod 700 ~/.ssh</code>	Restricts access to the SSH directory.
<code>chmod 600 ~/.ssh/authorized_keys</code>	Protects the <code>authorized_keys</code> file.
<code>sudo chown student:student hello.py</code>	Returns ownership of a file after it was edited with <code>sudo</code> .

Linux Navigation

Command	Purpose
<code>pwd</code>	Prints the current working directory.
<code>ls -l</code>	Lists files with permissions, ownership, and sizes.
<code>tree enauto -L 2</code>	Shows the folder structure of the virtual environment.

<code>mkdir -p ~/Projects/python</code>	Creates the project directory structure.
<code>cd ~/Projects/python</code>	Changes into the Python workspace.

Python Environment

Command	Purpose
<code>python3 --version</code>	Displays the installed Python version.
<code>pip3 --version</code>	Displays the installed pip version.
<code>python3 -m venv enauto</code>	Creates an isolated Python virtual environment.
<code>source enauto/bin/activate</code>	Activates the virtual environment.
<code>which python</code>	Shows which Python executable is currently being used.
<code>which pip</code>	Shows which pip executable belongs to the active environment.

Editing Files

Command	Purpose
<code>nano hello.py</code>	Creates or edits a Python file.
<code>Ctrl+O</code>	Writes (saves) the file in nano.
<code>Enter</code>	Confirms the filename when saving.
<code>Ctrl+X</code>	Exits nano.

Running Python

Command	Purpose
<code>python hello.py</code>	Executes a Python script.
<code>print("text")</code>	Displays text on the screen.
<code>type(variable)</code>	Returns the data type of a variable.

Packages Installed

Command	Purpose
<code>sudo apt install -y git curl wget vim nano tree htop unzip zip python3-pip python3-venv python3-dev build-essential net-tools dnsutils traceroute nmap</code>	Installs the core tools used for development, networking, troubleshooting, Python, and automation.

Key Concepts Learned

- Always update a fresh Linux installation first.
- Verify the system before installing software.
- Use SSH keys instead of passwords whenever practical.
- Work from a Python virtual environment for each project.
- Edit files in your home directory without sudo.
- Read Python errors carefully—they explain what went wrong.
- Build knowledge one layer at a time before moving into network automation.